

AAD - Mod 2

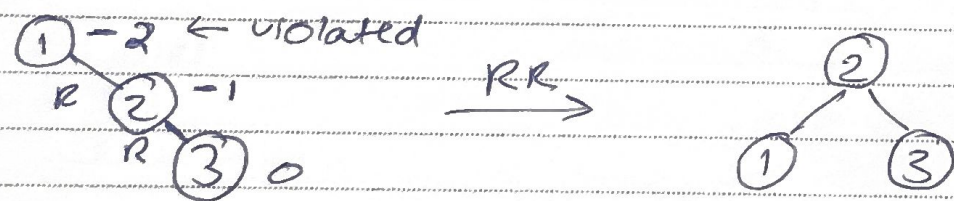
Self Balancing Tree

AVL Trees (Adelson-Velsky & Landis)

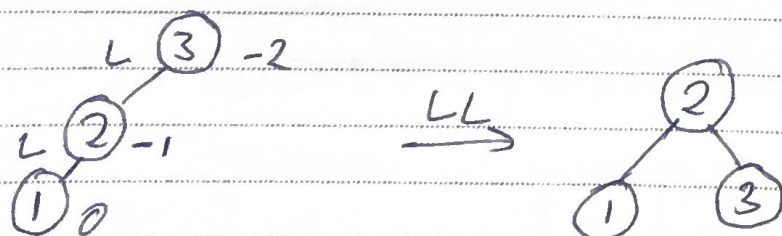
Insertion

Obtain balance factor = height of left subtree
- height of right subtree
= $\{-1, 0, 1\}$
↓
If any other values,

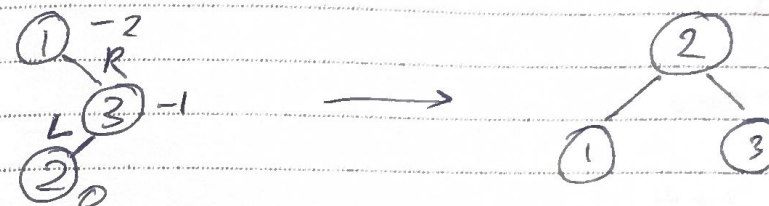
① RR → left rotate



② LL → right rotate

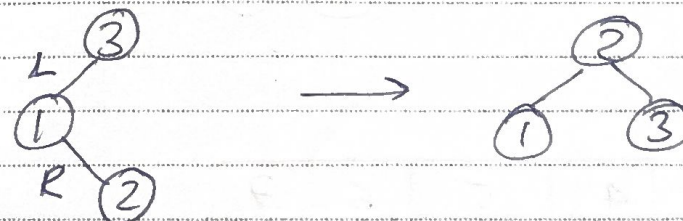


RL → Left then right.



[Hint: Find the middle value among the three and keep it as root, then follow the binary tree rule]

LR → Right then left



Deletion: Remove required node & balance with any of the above rotations at every step.

Disjoint Sets

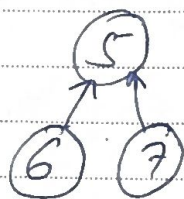
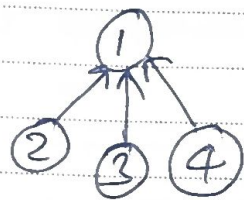
Disjoint sets

Representations

→ Linked lists

$$S1 = \{1, 2, 3, 4\}$$

$$S2 = \{5, 6, 7\}$$



→ Array

1	2	3	4	5	6	7
-4	1	1	1	-3	5	5

↓ ↓ no. of nodes
indication
of parent
node

↓ ↓
root node

Algorithm Make set(x)
Create new linked list

Algorithm make-set (x)

$n \rightarrow \text{data} = x$

$n \rightarrow \text{parent} = \text{NULL}$

Algorithm Find (n)

while $n \rightarrow \text{parent} \neq \text{NULL}$ do

$n = n \rightarrow \text{parent}$

return n

$T(n) = O(n)$

Algorithm Union (a, b)

$x = \text{Find}(a)$

$y = \text{Find}(b)$

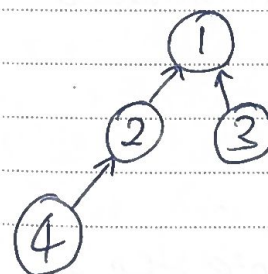
if $x \neq y$ then

$y \rightarrow \text{parent} = x$

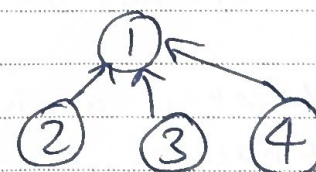
$T(n) = O(n)$

Path compression

Parent once found, directly link it with parent.



Find 4



NOVEMBER 2022

09

Wednesday

Week 46 ■ 313-052

OCTOBER							2022
Su	Mo	Tu	We	Th	Fr	Sa	
30	31						
2	3	4	5	6	7	8	1
9	10	11	12	13	14	15	2
16	17	18	19	20	21	22	3
23	24	25	26	27	28	29	4

Union by Rank (weighted rule)

Rank = depth

Union of two sets:

→ Same rank → Resulting set's rank is 1 larger

→ Diff → Resulting set's rank is larger of the two.

Graph traversal

Algorithm BFS (G, u)

- 1) Set all nodes are unvisited
- 2) Mark starting vertex u as unvisited & put it into empty Queue Q.
- 3) While Q is not empty
Dequeue v from Q
while v has an unvisited neighbour w

Mark w as visited

Enqueue w into Q

- 4) If there is any unvisited node x
Visit x & insert it into Q. Goto step 3.

$$T(n) = O(V+E)$$

DECEMBER 2022

Su	Mo	Tu	We	Th	Fr	Sa
					1	2
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

2022 NOVEMBER

Thursday

Week 46 ■ 314-051



10..

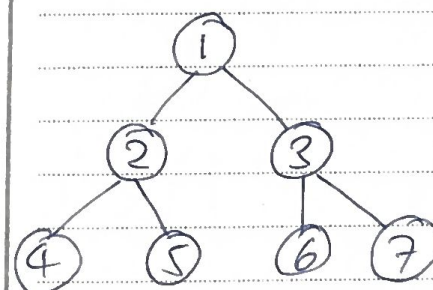
Algorithm DFS (G, u)

- 1) Mark vertex u as visited
- 2) For each adjacent vertex v of u
if v is not visited
DFS (G, v)

$$T(n) = O(V+E)$$

BFS: Visit all adjacent nodes before moving on to next. (Queue)

DFS: Visit one node, go to next and repeat until all nodes on that path are visited. Go back one step at a time if there are other unvisited nodes (Stack)



BFS: {1, 2, 3, 4, 5, 6, 7}

DFS: {1, 2, 4, 5, 3, 6, 7}

Topological sorting: Remove node with indegree zero, one-by-one.